

Hugo Miguel Miranda Barros, MS

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SUMMARY

AI Engineer with 5+ years bridging research and production, specialised in computer vision across 2D and 3D data. Background in bioengineering with hands-on experience across the full ML pipeline. Growing focus on GenAI/LLMs through independent study and personal projects.

SKILLS

ML/AI : CNNs, Transformers, RNNs, Reinforcement Learning, GenAI (LangGraph, OpenAI SDK), LLMs, Diffusion, NeRFs, classical **computer vision/image analysis**. Supervised and unsupervised learning. **OCR** (tesseract). **Full ML pipeline from data processing to deployment**

Coding : **Python** (PyTorch, Sklearn, Pandas, Matplotlib, NumPy), **C++**, **SQL** (basic), LaTeX

Workflow : **Grafana**, **WandB** (MLOps), **Confluence**, **Jira**, **Docker**, **Git**, **Ray** (distributed computing), **Flyte**, **Panderas**, **Poetry**, **uv**, ROS2

Cloud : **Azure** and **AWS** essentials (auto-scaling, distributed computing)

Languages : **Portuguese** (native), **English** (fluent), **Spanish** (essential)

EXPERIENCE

Ocean Infinity

Aug 2020 – Present

AI Researcher

- **Product Owner** for two projects from inception to MVP: quarterly planning, stakeholder engagement, and task distribution:
 - * Bayesian target search using **sonar** sensors: probability-based optimisation with a custom-built 2D simulation environment
 - * Real-time **alignment aid tool**: using segmentation of drill bit components to detect misalignment. Reduced 50% alignment time
- **Inspection**: segmentation and detection of assets, achieving **over 80% F1 score**, with complementary **OCR** (tesseract) for asset identification
- **Auto-annotation** product for object detection in **side-scan sonar images**, reducing data process time **over 50% from weeks to days**
- **Alignment** of multi-beam sonar surveys, correcting navigation drift through GPU-accelerated optimisation for xyz positioning, with over 90% time reduction in data processing, from weeks to hours
- **LLM-based classifier** for maritime Daily Progress Report generation, assigning 17 operational categories to time-stamped Excel event logs via structured-output prompting (LangChain + OpenAI), replacing manual report compilation
- **Multi-beam Point cloud** noise reduction and cleaning using networks such as PointNet, as well as Cube. Reduced the PC cleaning process from days to hours.
- **Autonomous pipeline following** for ROVs: fused magnetometer and navigation data with Kalman filtering to compute real-time cross-track corrections, replacing a joystick manual operator process
- Research into monocular and stereo **SLAM** using **NeRFs** and **Gaussian Splatting** for real-time navigation and 3D reconstruction; C++ integration via **ROS2** and FoxGlove
- **Classification and eventing** of underwater assets using CNNs and Transformers in video streams

Abyssal

Jun 2019 – Jul 2020

Research Intern & MSc Thesis

- **CNN prediction models** for monocular **depth estimation and 2D/3D segmentation**, using **GANs** for domain transfer from synthetic to real data. Preceded by an internship on CNN-based depth regression using simulated data

INESC TEC

Jan – Jun 2019

Research Intern

- Follicle detection and segmentation in 3D ovarian ultrasound images: comparative study of **classical methods** (Otsu, Canny, watershed, active contours) against a **CNN-based** approach

EDUCATION

Faculty of Engineering, University of Porto

2014 – 2020

MS, Bioengineering

- Thesis: Aquaculture Fish Quality Control Using Synthetic Data

AWARDS & CERTIFICATIONS

Merit Student Grant Award, Calouste Gulbenkian Foundation (2014) | **AWS Cloud Practitioner Essentials** (2020) | **15 Days of SQL: The Complete SQL Masterclass**, Udemy (2025)

EXTRA ACTIVITIES

Basic Life Support and Defibrillator Certified | Robotics Professor and Karate Instructor | Former member of BEST-Porto, European Youth Parliament, and VP of a Local Youth Political Party